Physical Chemistry

Science

May, 2026

Physical Chemistry (A11403)

Short Title:	Physical Chemistry		
Faculty	Science and Computing		
Department	Science		
Cluster	Chemistry		
Author	EOWENS		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module will discuss the molecular theory of gases, the behaviour of real gases, colligative properties of solutions, vapour liquid equilibria, aqueous acid base equilibria, solubility and complexation equilibria, kinetics, and equilibria and chemical thermodynamics.

Programmes

CHEM-0008	BSc (Hons) in Pharmaceutical Science (WD_SPHSC_B)	2 / 3 / M
CHEM-0008	BSc in Pharmaceutical Science (WD_SPHSC_D)	2 / 3 / M

Indicative Content

- Kinetic molecular theory of gases and the properties of real gases
- Colligative properties of solutions and vapour liquid equilibria
- The first and second laws of thermodynamics; the criterion for spontaneity; Gibbs free energy and its relationship to the equilibrium constant
- Chemical kinetics and equilibrium; experimental methods for determining rate laws; temperature dependence on rate laws; radiochemical dating and pharmaceutical shelf-life
- Aqueous acid base equilibria
- Solubility and complexation equilibria

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the behaviours of a real and an ideal gas, use the van der Waals equation, and relate the van der Waals coefficients to molecular structure. Apply the equation for root mean speed, mean free path and Maxwell-Boltzman distribution.
2. Apply the integrated rate laws and half-life equations for zero, first and second order rate equations. Determine rate laws for zero, first and second order reactions using different experimental methodologies.
3. Apply the Arrhenius equation to determine the relationship between temperature and reaction rate.
4. Calculate the change in internal energy, enthalpy and Gibbs free energy of a system. Identify the criterion for spontaneity.
5. Determine Ka, Kb, Ka, pKb, and describe its relationship between strength and molecular structure.
6. Determine the solubility of an ionic compound and describe the factors which affect its solubility.
7. Use a range of techniques to examine the kinetics, energy changes and equilibrium constant of a system.
8. Communicate by written and verbal means the results of theory and practical-based exercises.

Learning and Teaching Methods

- Lectures.

- Practicals.
- Tutorials.
- Self-directed study.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1,2,3,4,5,6
Continuous Assessment	50%	
Practical	50%	7,8

Assessment Criteria

<40%: Inability to demonstrate an understanding of the basic concepts of physical chemistry.

40%-49%: Able to show a basic knowledge of the concepts outlined in the indicative content. Able to provide partial solutions to numerical problems and have some degree of awareness as to the importance and correct use of dimensions for physical quantities.

50%-59%: The student will be able to demonstrate a considerable understanding of the complexities of the topics listed in the indicative content and apply suitable problem-solving techniques to obtain numerical solutions.

60%-69%: In addition to the above, the student will demonstrate the ability to apply problem-solving techniques to new similar problems and demonstrate a good understanding of the majority of the course content.

70%-100%: Will be able to demonstrate comprehensive knowledge and understanding of all learning outcomes, as well as a superior standard of reporting and presentation skills.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the final exam and continuous assessment components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Tutorial	6	
Practical	30	
Independent Learning	75	

Essential Material

- Atkins, P. and J. De Paula. _The Elements of Physical Chemistry_. 6th ed. Oxford: Oxford University Press, 2013.
- Brown, T., H. Lemay, B. Burnsten and C. Murphy. _Chemistry: The Central Science_. 3rd ed. Australia: Pearson, 2013.

Supplementary Material

- "WIT." www.wit.ie/library/webct. www.wit.ie/library/webct

Requested Resources

- Room Type: Chemistry Lab